DRAFT STANDARD OPERATING PROCEDURE

Germplasm Passport Data Management

Sugarcane Breeding and Germplasm Conservation Program

SOP No.	SCB-GC-SOP-01	**Revision**	0
Effective date	_____	**Supersedes**	New
Prepared by	_____	**Reviewed by**	_____
Approved by	_____	**Controlled copy**	Yes / No

Status: Draft technical SOP. Field dimensions, thresholds, chemical treatments, quarantine steps, and local compliance requirements must be validated and approved by the implementing institution before adoption.

1. Purpose

To establish a permanent, traceable, standardized identity record for every sugarcane accession acquired, collected, exchanged, conserved, distributed, characterized, or used in breeding. The protocol separates passport information (identity and origin) from characterization and evaluation data so that accession history remains auditable across field, in-vitro, molecular, and breeding records.

2. Scope

Applies to Saccharum species, commercial hybrids, interspecific and intergeneric materials, breeding parents, genetic stocks, wild collections, imported germplasm, and locally collected accessions maintained by the program.

3. Core rule

One accession receives one permanent accession number. The permanent number must never be reassigned or reused, even if the accession is later lost, declared duplicate, renamed, or removed from the active collection.

4. Responsibilities

- Germplasm curator: approves accession creation, verifies source documentation, resolves synonyms and status changes.
- Breeder/collector: supplies pedigree, collection, donor, and trait-use information and signs the intake record.
- Database custodian: maintains controlled vocabulary, backups, audit trail, and record-linkage to field and laboratory samples.
- Plant health/quarantine staff: records phytosanitary and release status before the material enters the active collection.

5. Required records and tools

- Accession intake form and donor/collection documentation.
- Central electronic germplasm database with routine backup.
- Permanent field labels plus barcode/QR code where available.
- GPS-enabled device for new field collections.
- Photographic record and chain-of-custody log.
- Linked IDs for tissue-culture, DNA, herbarium/reference sample, and safety-duplicate holdings where used.

6. Procedure

1. Receive material in a designated intake or quarantine area. Do not plant directly into the active germplasm field until plant-health requirements are satisfied.
2. Reconcile all documents against the physical shipment: accession/variety name, donor number, source institution, country, pedigree, quantity, material type, and phytosanitary documentation.
3. Assign the next permanent institutional accession number. Recommended format: [institution]-GC-[sequential number], e.g., SRA-GC-0001. Never encode mutable traits such as taxonomic status or field location in the permanent number.
4. Enter minimum passport fields using FAO/Bioversity MCPD v2.1 terminology wherever possible. Preserve original donor spelling and identifiers in dedicated source fields.
5. Record synonyms and previous numbers as cross-references; do not overwrite the permanent accession ID.
6. For new collections, record latitude, longitude, elevation, collection date, site description, habitat, population size, and collector information at the time of collection. If a coordinate is unknown, leave it unknown rather than estimating false precision.
7. Assign current conservation status and location codes (field, in-vitro, cryo, DNA, safety duplicate, quarantine, lost, inactive). Every movement between fields or stores must be recorded.
8. Create standardized photographs showing label plus representative plant/stalk material and link them to the accession record.
9. Perform identity verification at establishment and whenever a label conflict, off-type, synonym issue, or suspected mixture occurs. Use morphology, pedigree/source records, photographs, and molecular fingerprints when available.
10. Complete a scheduled database audit at least annually and after every major field regeneration.

7. Minimum passport fields

Data item	MCPD-aligned field	Status	Guidance
Permanent accession number	ACCENUMB	Required	Institutional unique identifier
Persistent unique identifier/DOI	PUID	If assigned	Global persistent identifier
Holding institution	INSTCODE	Required	Institution code/name
Accession/variety name	ACCENAME	Required if known	Preserve source spelling
Genus/species	GENUS / SPECIES	Required where known	Taxonomic identity
Country of origin	ORIGCTY	If known	Use standard country code where possible
Collection number	COLLNUMB	For collected material	Original collector number
Collection site	COLLSITE	For collected material	Locality/site description
Latitude/Longitude	DECLATITUDE / DECLONGITUDE	For collected material	Decimal degrees preferred
Elevation	ELEVATION	If known	Meters above sea level
Collection date	COLLDATE	If applicable	Record original date
Acquisition date	ACQDATE	Required	Institutional receipt/accession date
Biological status	SAMPSTAT	Required	Wild, landrace, breeding/research, etc.
Pedigree/ancestry	ANCEST	For breeding material	Female x male; generation where known
Donor institution	DONORCODE / DONORNAME	If donated	Source organization
Donor accession no.	DONORNUMB	If supplied	Do not replace institutional ID
Safety duplicate site	DUPLSITE	If held	Backup location
Storage/conservation	STORAGE	Required	Field, in-vitro, cryo, seed, DNA, etc.
Remarks	REMARKS	As needed	Identity issues, restrictions, historical notes

8. Sugarcane-specific supplementary fields

- Female parent, male parent, cross number, cross year, and breeding series.
- Breeder and crossing station.
- Original selection number and year first selected.
- Germplasm class: Saccharum species, commercial hybrid, introgression/pre-breeding material, allied genus, genetic stock.

- Flowering behavior, known use as male/female parent, and major breeding-use notes (kept separate from formal evaluation data).
- Current health/identity flag: normal, identity under verification, mixed/contaminated, quarantine, lost, inactive.

9. Quality-control checkpoints

- No duplicate permanent IDs.
- Field label, database record, and sample labels agree before planting or distribution.
- Source identifiers are preserved and not silently normalized away.
- Corrections retain an audit trail: old value, new value, date, reason, and responsible person.
- A safety duplicate is documented for irreplaceable or high-value accessions when resources permit.
- Backups are tested for recoverability, not merely created.

10. Accession intake / audit form

Field	Entry
Permanent accession number	
Accession/variety name	
Donor / collector	
Donor number	
Country and locality of origin	
Latitude / longitude / elevation	
Pedigree / ancestry	
Acquisition date	
Material received	
Plant-health/quarantine status	
Primary conservation location	
Safety duplicate location	
Synonyms / other numbers	
Identity verification method	
Curator remarks	
Verified by / date	

References and technical basis

FAO/Bioversity Multi-Crop Passport Descriptors v2.1 (2015) - Web source

ICAR-Sugarcane Breeding Institute Research Centre, Kannur - world sugarcane germplasm collection - Web source